Best-first node selection Margin Preservation under Bounded Score Perturbations

Research Architecture Runtime

`operator/best-first-node-selection`

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Abstract

`best-first-node-selection` selects a unique maximizer. Margin $\gamma(s) > 2\varepsilon$ is necessary and sufficient for unique-max invariance under $\|\delta\|_\infty \leq \varepsilon$. Lean `LEAN_FULL` (reduction to `Argmax.Margin`).

This theorem is: `Reduction`

1 Problem

The `best-first-node-selection` operator returns the unique maximizer index $i^\star$ of scores $s \in \mathbb{R}^m$ ($m \geq 2$), up to the operator's definitional presentation (heap top, tournament winner, masked max, etc.).

2 Stability notion

Perturbations: $\|\delta\|_\infty \leq \varepsilon$. Let $\gamma(s) = s_{i^\star} - \max_{j \neq i^\star} s_j$. Stability: $\gamma(s) > 2\varepsilon$ implies $i^\star$ remains the unique maximizer of $s + \delta$. Sharpness: $\gamma(s) \leq 2\varepsilon$ admits an adversary destroying unique maximality.

3 Definitions

Definition 1 (Unique maximizer). $i^\star$ uniquely maximizes s if $s_j \leq s_{i^\star}$ for all j and $s_j = s_{i^\star} \Rightarrow j = i^\star$.

Definition 2 (Margin). $\gamma(s) = s_{i^\star} - \max_{j \neq i^\star} s_j$.

4 Theorem

Theorem 3 (Margin invariance). *If $i^\star$ uniquely maximizes s and $s_{i^\star} - s_j > 2\varepsilon$ for all $j \neq i^\star$, then for every $\|\delta\|_\infty \leq \varepsilon$, $i^\star$ uniquely maximizes $s + \delta$.*

Theorem 4 (Sharpness). *If some $j \neq i^\star$ has $s_{i^\star} - s_j \leq 2\varepsilon$, an admissible δ destroys unique maximality.*

5 Intuition

Depressing the winner by ε and elevating a rival by ε consumes 2ε of margin. Strict margin above 2ε leaves the winner unique; otherwise a two-point adversary forces a tie or loss.

6 Examples

Example 5. Scores $(7, 5, 4)$, $i^\star = 0$, $\gamma = 2$. For $\varepsilon = 0.8$, $2\varepsilon = 1.6 < 2$, so the winner is stable. For $\varepsilon = 1$, $2\varepsilon = 2 \geq \gamma$; $\delta = (-1, +1, 0)$ yields a tie between the first two scores.

7 Proof

Let $s' = s + \delta$ with $\|\delta\|_\infty \leq \varepsilon$. For $j \neq i^\star$,

$$s'_{i^\star} - s'_j = (s_{i^\star} - s_j) + (\delta_{i^\star} - \delta_j) \geq (s_{i^\star} - s_j) - 2\varepsilon > 0,$$

using $s_{i^\star} - s_j > 2\varepsilon$. Thus $s'_j < s'_{i^\star}$ for all $j \neq i^\star$, so $i^\star$ uniquely maximizes s' (Theorem **??**).

For sharpness, pick j with $s_{i^\star} - s_j \leq 2\varepsilon$, set $\delta_{i^\star} = -\varepsilon$, $\delta_j = +\varepsilon$, and 0 elsewhere. Then $\|\delta\|_\infty \leq \varepsilon$ and $s'_{i^\star} - s'_j = \gamma_{ij} - 2\varepsilon \leq 0$, so unique maximality fails (Theorem **??**).

8 Formal statement

Lean module

`Research.Operators.BestFirstNodeSelection.Margin`

Source file

`lean/Research/Operators/BestFirstNodeSelection/Margin.lean`

Theorems

- `best\_first\_node\_selection\_margin\_invariance`
- `best\_first\_node\_selection\_margin\_sharpness`

Note

Definitional reduction of Argmax.Margin.

Certificate

`lean/certificates/best-first-node-selection/best-first-node-selection-margin`

Status

`LEAN_FULL` on Mathlib $\mathbb{R}$ (`REAL_MATHLIB`)

9 Proof dependencies

- `Research.Operators.Argmax.Margin` (`margin\_invariance`, `margin\_sharpness`).
- ℓ_∞ gap shrinkage by at most 2ε.
- No other operator theorems required.

10 Consequences

Any unique-max presentation of `best-first-node-selection` inherits Argmax margin stability. Related: `argmax`, `tournament-winner`, `greedy-maximum-selection`.